

Multiple Choice (10 marks)

1. What is the least perfect square with 3 different prime factors?
 - (a) 900
 - (b) 441
 - (c) 300
 - (d) 484
 - (e) 529
2. Let M be a 5×5 real matrix. Exactly four of the following five conditions on M are equivalent to each other. Which of the five conditions is equivalent to NONE of the other four?
 - (a) The column vectors of M form a basis for R^5 .
 - (b) For any two distinct column vectors u and v of M , the set $\{u, v\}$ is linearly independent.
 - (c) The trace of M is nonzero.
 - (d) The eigenvalues of M are all nonzero.
 - (e) The homogeneous system $Mx = 0$ has only the trivial solution.
3. Find all positive values of c so that the inequality $x^2 - 6x + c < 0$ has real solutions for x . Express your answer in interval notation.
 - (a) $(0, 0)$
 - (b) $(6, 0)$
 - (c) $(9, 16)$
 - (d) $(0, 9)$
 - (e) $(3, 6)$
4. Which pair of expressions below are equivalent?
 - (a) $2x + 2x$ and $2x^2$
 - (b) $x+x+x+x$ and x^4
 - (c) $3x + 5x$ and $15x$
 - (d) $4(2x - 6)$ and $8x - 24$
 - (e) $3x + 3x + 3x$ and $3x^3$
5. What is the probability that a randomly selected integer in the set $\{1, 2, 3, \dots, 100\}$ is divisible by 2 and not divisible by 3? Express your answer as a common fraction.

- (a) $\frac{17}{31}$
- (b) $\frac{33}{66}$
- (c) $\frac{33}{100}$
- (d) $\frac{17}{50}$
- (e) $\frac{1}{6}$

True / False (10 marks)

Answer True or False

6. True or False: The sum of 4 over 7 and 2 over 7 is equal to 6 over 7.
7. True or False: If $_{42}$ baps are converted to daps, the total equals 40 daps based on the given relationships between the units.
8. True or False: The area of the region between the graphs of $f(x) = x^2 - 4x + 10$ and $g(x) = 4x - x^2$ from $x = 1$ to $x = 3$ is 2.987.
9. True or False: 26.1 mm is equal to 0.0261 dm, which incorrectly applies the conversion factor.
10. True or False: The value of y that satisfies the similarity condition for the polygons is 3.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. The graph of $y^2 + 2xy + 40|x| = 400$ partitions the plane into several regions. What is the area of the bounded region?
12. Given the function $h(x) = (x^{\frac{1}{2}} - 1/2 + 2x)(7 - x^{\frac{1}{2}} - 1)$, derive $h'(x)$ and evaluate it at $x = 4$. Discuss the significance of your findings in the context of the function's behavior.

End of Paper